

2.1 Cell Metabolism I

▼ Basics/Warm-Up

▼ What is metabolism?

- Metabolism describes the key processes that occur within our bodies that keep us alive

▼ What is a catabolic reaction?

- A reaction that involves the breakdown of a larger molecule into smaller ones
- It typically releases energy which is stored within ATP

▼ What is an anabolic reaction?

- A reaction that involves the formation of a larger molecule from smaller ones.
- It typically requires the input of energy from ATP

▼ What is the cytosol?

- The liquid part of the cytoplasm in a cell.

▼ Where does the term glycolysis originate from?

- Glyco (sugar) lysis (breaking)
- Glycolysis is hence the breakdown of sugar (glucose) into two molecules of pyruvate.



▼ The six main reaction types.

Just 6 types of reactions define metabolism

Type of Reaction	Description
Oxidation-reduction	Electron transfer
Ligation requiring ATP cleavage	Formation of covalent bonds (i.e., carbon-carbon bonds)
Isomerization	Rearrangement of atoms to form isomers
Group transfer	Transfer of a functional group from one molecule to another
Hydrolytic	Cleavage of bonds by the addition of water
Addition or removal of functional groups	Addition of functional groups to double bonds or their removal to form double bonds

▼ What are the three main steps of cellular metabolism?

- Glycolysis
- Krebs Cycle / TCA Cycle
- Oxidative Phosphorylation

▼ Which reactions occur in the cytosol/mitochondria?

- Glycolysis in the cytosol
- TCA Cycle and Oxidative Phosphorylation in the mitochondria

▼ Where do the urea and water produced from the reactions in aerobic respiration originate from?

- Water from the reduction of oxygen
- Urea from the breakdown of amino acids

▼ How much energy is released from the combustion of one mole of glucose?

- ~2872 kJ

▼ How can we get this energy to be released from glucose?

- Through the **complete oxidation (aerobic respiration) of glucose**
- This reaction requires energy to proceed and reach the activation energy before proceeding

▼ How is the activation energy for the complete combustion of glucose reached?

- Using **heat energy** from the body's temperature or **enzymes**

▼ How many molecules of ATP can be synthesised from the complete oxidation of glucose?

- 36-38.

▼ What is ligation?

- The joining of two **nucleic acid** fragments **through the action of an enzyme**.

▼ What is a kinase?

- A kinase is an enzyme that always catalyses the **addition of phosphate groups** to other molecules

▼ What happens in the first stage of glycolysis?

- Glucose $\rightarrow$ Glucose-6-phosphate
- ATP $\rightarrow$ ADP + P_i
- Catalysed by **hexokinase**

▼ What happens in the second stage of glycolysis?

- **Glucose-6-phosphate** changes to **fructose-6-phosphate** in an isomerisation reaction
- Catalysed by **phosphoglucose isomerase**.

▼ What is the purpose of this second stage of glycolysis?

- It happens because **fructose can split into equal halves when cleaved**
- Glucose, on the other hand, cannot

▼ What happens in the third stage of glycolysis?

- **Fructose-6-phosphate** $\rightarrow$ **Fructose 1,6-bisphosphate**
- ATP $\rightarrow$ ADP + P_i
- Catalysed by **phosphofructokinase**

▼ What happens in the fourth stage of glycolysis?

- **Fructose-1-6-bisphosphate** $\rightarrow$ **Glyceraldehyde-3-phosphate (GALP) + Dihydroxyacetone phosphate (DHAP)**
- Catalysed by **aldolase**

▼ What happens in the fifth stage of glycolysis?

- **Dihydroxyacetone phosphate** $\rightarrow$ **Glyceraldehyde-3-phosphate**
- Catalysed by **triose phosphate isomerase (TPI)**

▼ What happens in the sixth stage of glycolysis?

- **Glyceraldehyde-3-phosphate** → **1,3-bisphosphoglycerate**
- $\text{NAD}^+ \rightarrow \text{NADH} + \text{P}_i$
- Inorganic phosphate is also added to glyceraldehyde-3-phosphate, to prepare it for substrate-level phosphorylation of two molecules of ATP, not one.
- Catalysed by **glyceraldehyde-3-phosphate dehydrogenase**
- This enzyme needs NAD^+ to function, hence glycolysis is reliant on the presence of NAD^+ to go ahead, otherwise this step will not proceed and pyruvate won't be produced.

▼ What happens in the seventh stage of glycolysis?

- **1,3-bisphosphoglycerate** → **3-phosphoglycerate**
- $\text{ADP} + \text{P}_i \rightarrow \text{ATP}$
- Catalysed by **phosphoglycerate kinase**

▼ What happens in the eighth stage of glycolysis?

- **3-phosphoglycerate** → **2-phosphoglycerate**
- Catalysed by **phosphoglycerate mutase**

▼ What happens in the ninth stage of glycolysis?

- 2-phosphoglycerate → Phosphoenolpyruvate + H_2O
- Catalysed by **enolase**

▼ What happens in the tenth stage of glycolysis?

- Phosphoenolpyruvate → Pyruvate
- $\text{ADP} + \text{P}_i \rightarrow \text{ATP}$
- Catalysed by **pyruvate kinase**

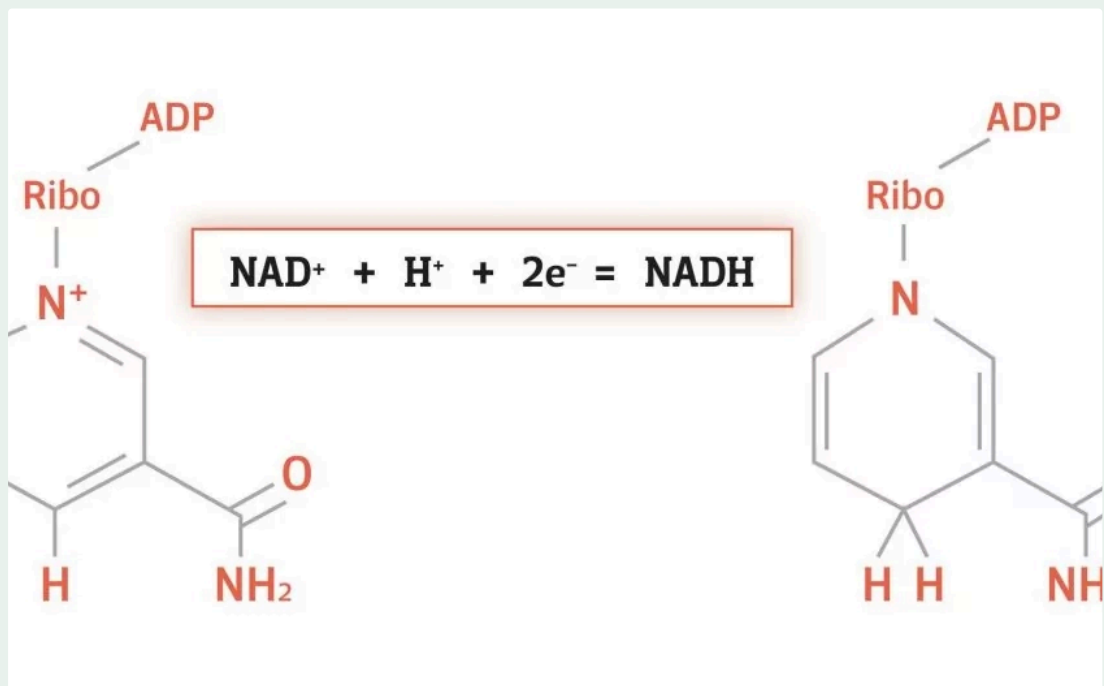
▼ What are the net products of glycolysis?

- **2x pyruvate**
- **2x NADH**
- **2x ATP**, 4x produced by 2x pyruvate, but 2 were used initially to form glyceraldehyde-3-phosphate.

► The deficiency of which glycolytic enzyme can prove to be fatal in humans and why?

► What is the purpose of adding a phosphate group to a glucose molecule in the first step?

- ▶ How does the addition of this phosphate group allow glucose to be committed to metabolism?
- ▶ What do dehydrogenase enzymes need to function?
- ▼ Describe briefly what the pentose-phosphate pathway is.
 - It is a parallel pathway to glycolysis used to **generate biosynthetic molecules from substrates of glycolysis**
 - For example, glucose-6-phosphate can be converted to ribose-5-phosphate which is a key component for RNA/DNA.
 - **NADPH** comes from this pathway too. (Results from NADH)
- ▼ What is the difference between NADPH and NADH?
 - **NADPH** has an **additional phosphate group** and is typically used for **anabolic reactions**
- ▼ Which molecule, produced via the pentose-phosphate-pathway is vital to maintaining the integrity of RBCs?
 - **NADPH**
- ▶ Describe how NADPH maintains the integrity of RBCs.
- ▼ What is the difference between NADH and NAD⁺?
 - **NADH** and **NAD⁺** are forms of the same molecule
 - **NADH** is the **reduced form**, **NAD⁺** is the **oxidised form**



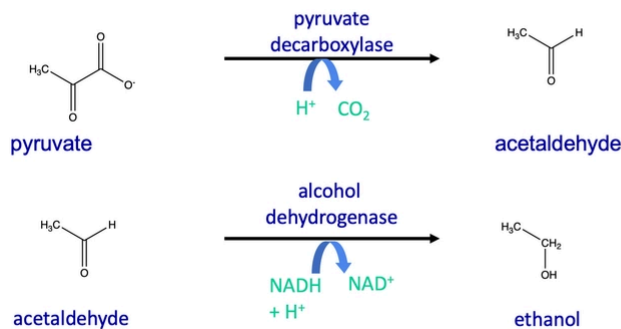
▼ What are the three fates of pyruvate?

- **Alcoholic fermentation** - yeast
- **Lactate production** - human anaerobic
- **Acetyl CoA production** - human aerobic

▼ What reactions occur in alcoholic fermentation?

- **Ethanol** is ultimately produced from pyruvate in a two-stage reaction

Pyruvate has three possible fates



(i) Alcoholic Fermentation

- This is characteristic of **yeasts** and can occur under **anaerobic** conditions.

▼ What role do alcoholic fermentation and lactate production play in cell metabolism?

- **Regenerating NAD+**
- The **lactate** and **ethanol** produced by these reactions are "waste" products, the main aim of the reactions is to regenerate NAD+ from NADH to keep ATP production going
- This is mainly important in **anaerobic conditions** as **NADH is not useful to us** in states of **hypoxia**, as it is used in oxidative phosphorylation, so regeneration of NAD+ allows for **ATP production via glycolysis**
 - NAD+ can be used in the **6th step of glycolysis** (glyceraldehyde-3-phosphate → 1,3-glyceraldehyde biphosphate) which can subsequently help us to produce ATP in the remaining reactions of glycolysis

▼ How does creatine phosphate function in metabolism?

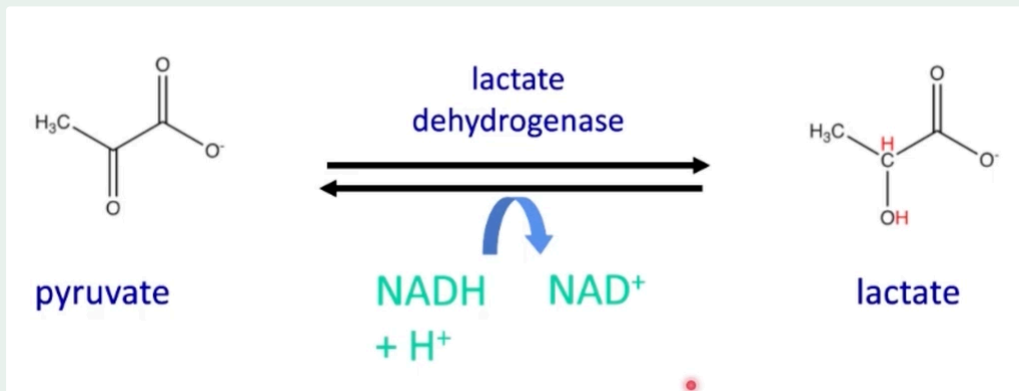
- Creatine phosphate has one phosphate group removed from it to return to its original creatine state and **phosphorylate the ADP to form ATP**
- It is used as ATP is usually in high demand during exercise + muscle contractions and so it **acts as a reservoir** for these demanding conditions

▼ What is the acetyl CoA produced by pyruvate used for?

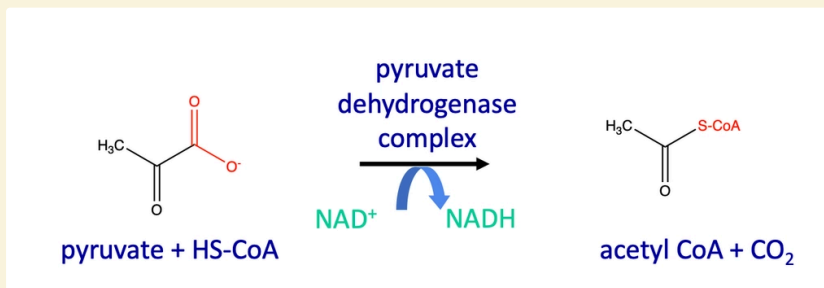
- The TCA Cycle.

▼ What is the role of the enzyme lactate dehydrogenase?

- Catalyses the **reduction of pyruvate to lactate** and vice-versa
- This reaction regenerates NAD^+ which allows glycolysis to continue (step 6)
 - NADH is not useful to us in states of hypoxia, as it is used in oxidative phosphorylation, so regeneration of NAD^+ allows for ATP production in glycolysis



▼ Describe how pyruvate is converted to acetyl CoA.



- $\text{NAD}^+ \rightarrow \text{NADH}$
- HS-CoA is just coenzyme A
- The **pyruvate dehydrogenase complex** catalyses this reaction

What is Peri Peri?